

chip required to enable augmented reality in sunglasses will be a lot smaller than that of, for example, a mobile phone. Miniaturisation is becoming increasingly feasible, thanks to micro-electromechanical systems (MEMS) and the development of cost-efficient batch fabrication techniques for their manufacture. MEMS are making it possible to integrate components such as microprocessors, sensors and radio communication circuits into a single integrated circuit (IC) or system-on-chip (SoC).

Open source platforms will play a key role in developing and deploying wearable technology, just as in the IoT, for much the same reasons. Only if the data from the device is accessible and users are able to harness it using multiple applications, will it be attractive in the long run. If the data from a device is always to be used only with a proprietary app, it will not be so much fun after a while!

3D printing—everything from spare parts to model planes

Trend: 3D printing has given wings to the dreams of many a do-it-yourself enthusiast! Interestingly, the cost of 3D printers has dropped significantly in the past few years. Previously, professional 3D printers used to cost hundreds of thousands of dollars, putting them out of the reach of individuals, hobby groups and colleges. However, now it is possible to own 3D printers like the MakerBot Replicator 2 for about \$2,200.

The viability of 3D printing has emerged in tandem with the advancement and democratisation of 3D design or CAD software, which allows users to easily create a digital 3D model of an object and optimise the design before anything has been built. The combination of these two technologies—3D design software and 3D printers—means that it is easier for individuals to convert an idea for a product into a physical object.

Trend+: There is a lot of open source activity in the 3D printing space. From design software and product designs, to the design of printers themselves, a lot of material is open sourced. RepRap, for example, is an open source self-replicating 3D printer, using which you can 3D print the parts and make more printers! According to the team, “RepRap was the first of the low-cost 3D printers, and the RepRap Project started the open source 3D printer revolution. It has become the most widely-used 3D printer among the global members of the Maker Community.”

A more recent innovation that has been open sourced, following the path of RepRap, is a 3D metal printer unveiled by Michigan Technical University in December 2013. Unlike current 3D printers that use plastic and similar materials for printing, this sub-\$1500 machine can print sturdy metal parts. The parts used in the device include a small commercial MIG welder and an open source microcontroller. According to news reports, the

printer is still a work-in-progress, but the team believes that it will be perfected very soon, once the vibrant open source community gains momentum. Everything required to build the printer, including detailed plans, software and firmware, is freely available.

Likewise, there is no dearth of 3D design tools such as FreeCAD, Wings3D, Seamless3D, and such. You can also find countless 3D designs to print on the Web, ranging from 3D greeting cards and toys to model airplanes. As a starting point, you could look up Makerbot or Thingiverse.

The more the merrier, in gaming too

Trend: Gaming is serious business these days. There is a lot of activity in this space, and people are spending big money on it! Understandably, open source principles and technology have pervaded gaming too. From open source games, to game design tools and console designs, there are loads of open source resources for game makers and players too. Open source games, generally made using open tools and open source software, are available not just for the Linux platform but for Windows, Mac, Arduino and other platforms too. While some games are just-for-fun, others are made with serious goals, such as sending across a serious social message or designing a serious strategy game for a well-defined target audience.

Trend+: Generally, making a game calls for selecting a game engine, a graphics engine, an AI framework, graphics tools, game design tools, sound tools, version control, project tracking, development IDE, and packaging and build tools. All these are available in the open source repertoire. In a wonderful article, Casper Bodewitz explains how he chose open source tools for each of these categories when building the strategy game Tenshu General (http://www.gamasutra.com/blogs/CasperBodewitz/20130913/198043/Open_source_tooling_try_it_before_you_buy_it.php). There are also full-fledged game development tool chains and cloud-based platforms.

One interesting tool for 3D game development is Panda3D, developed by Disney and maintained by Carnegie Mellon Entertainment Technology Centre. The Panda3D game engine is a framework for 3D rendering and game development for Python and C++ programs. It is open source, and can be used for even commercial ventures due to the liberal terms of its licence.

Ogre is another notable open source 3D graphics rendering engine for making fantastic games. Other good open source game engines to test would be Object-Oriented Graphics Rendering, Quake 3, Crystal Space and Irrlicht Engine. 

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